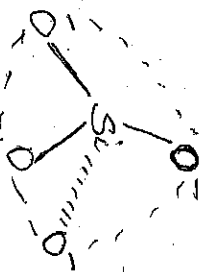


SILICATES

A silicate is any compound family of anions that consist of ~~silicon~~ ^{silicon} and oxygen often represented by the formula SiO_4^{4-} . These anions are very large polymeric molecules with many varieties of structures that include the rings, chains, as well as sheets. The several patterns of linkage is depending on the way the tetrahedral units are linked. The general classification of the silicates is:

- (I) Orthosilicates (or nesosilicates)
- (II) Pyrosilicates (or sorosilicates)
- (III) Cycle silicates (or Ring silicates)
- (IV) chain silicates (or pyroxenes)
- (V) Double-chain silicates (or amphiboles)
- (VI) Sheet silicates (or phyllosilicates)
- (VII) Three-dimensional (or tecto silicates)

→ The orthosilicates are the basic or simplest silicates that contain discrete SiO_4^{4-} tetrahedral units. They are structurally shown as below:



side view.

The orthosilicates ion is the strong conjugate base of weak orthosilicic acid as it will not persist in aqueous solution, hence orthosilicate minerals are rare in nature. They are only found with cations which form highly insoluble salts.

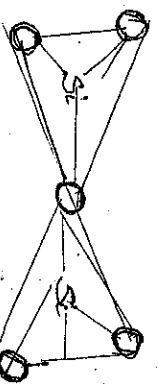
Common examples of orthosilicates are:
 1, Phenacite or Phenakite — Be_2SiO_4

2, Willemite — Zn_2SiO_4 ; This is a min of silicate ore of Zn. It is highly fluorescent (green) under UV-light.
 NB: in the above e.g.s, the Be^{2+} and Zn^{2+} ions are tetrahedrally surrounded by the oxygen atoms of the silicate in the structure.

3, Olivine — $(\text{FeMg})_2\text{SiO}_4$; It is green in colour, but the cations are octahedrally coordinated to the oxygen atoms of the silicate.

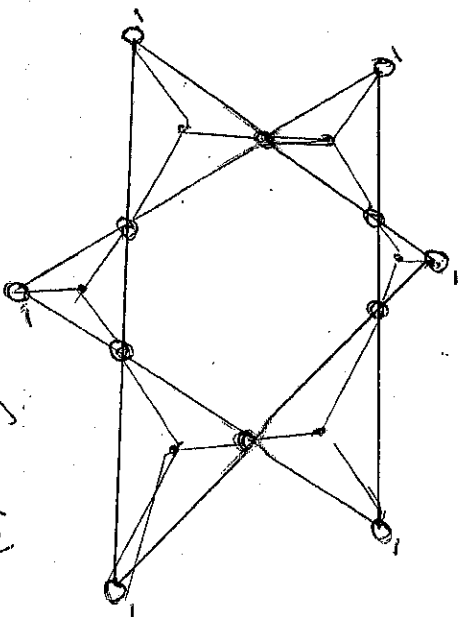
4, Zircon — ZrSiO_4 ; This is the oldest mineral on earth. The coordination number of Zr^{4+} is 8.

$\Rightarrow$ The pyrosilicate contains $\text{Si}_2\text{O}_7^{6-}$ ions that are formed by joining two tetrahedral SiO_4^{4-} which share one oxygen atom at one corner. They are structurally represented as below:



The pyrosilicate ion $\text{Si}_2\text{O}_7^{6-}$ is less basic than the orthosilicate ion SiO_4^{4-} . The only mineral in nature that contains the pyrosilicate ion is Thortveitite — $\text{Sc}_2\text{Si}_2\text{O}_7$ — hence the only known example of the silicate.

$\Rightarrow$ The cyclic silicates also called ring silicates contain the ion $(\text{SiO}_3)_n^{2n-}$ that are formed by linking three or more tetrahedral units SiO_4^{4-} cyclically. Each unit shares two oxygen atoms with the next units as shown below:

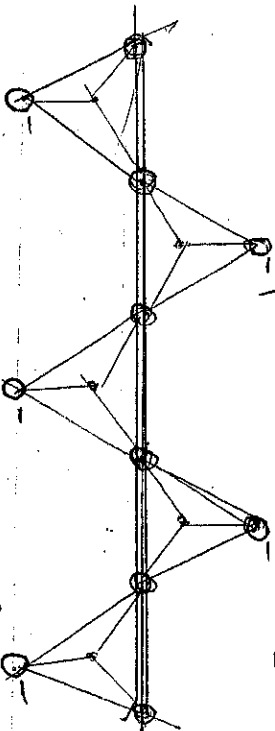


Common examples of the silicate are;
1/ Benitoite - $\text{BaTi}(\text{SiO}_3)_2$: containing three tetrahedra arranged cyclically i.e. $[\text{Si}_3\text{O}_9]^{6-}$.

2/ Beryl - $\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$: containing six-silicate rings $(\text{Si}_6\text{O}_{18})^{12-}$. It is an aluminosilicate where each aluminium is surrounded by 6-oxygen atoms octahedrally. well known varieties of beryl include Emerald and Aquamarine.

$\Rightarrow$ The chain silicates or pyroxenes contain $(\text{SiO}_3)^{2n-}$ ions that are formed by linking 'n' number of tetrahedral SiO_4^{4-} units linearly. Each unit shares two oxygen atoms with other units.

The structure of chain silicates could be shown as:



Common examples of chain silicates are;

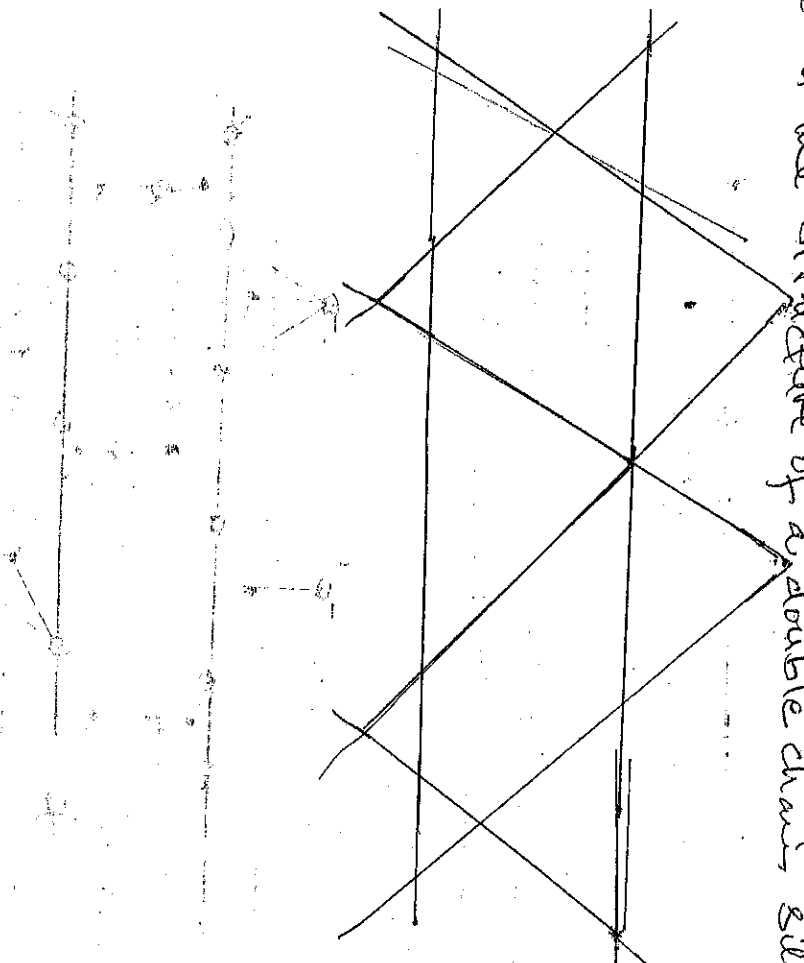
(1) Spodumene - $\text{LiAl}(\text{SiO}_3)_2$ - a pyroxene mineral consisting of lithium aluminium inosilicate.

(2) Diopside - $\text{CaMg}(\text{SiO}_3)_2$

(3) Wollastonite - $\text{Ca}_3(\text{SiO}_3)_3$

⇒ The double chain silicates or Amphiboles have the general formula $(\text{Si}_4\text{O}_{11})_n^{n-}$. Here there are two types of tetrahedra, the 6O sharing 3 vertices and the 8O sharing only 2 vertices.

Below is the structure of a double chain silicate.



Common examples of this type silicate are: Amphibole silicate

(1) Asbestos - These are non-combustible fibrous silicates. These have been used for thermal insulation, brake linings, construction material and filters. They are carcinogenic hence, have restricted or controlled applications.

(2) Tremolite - $\text{Ca}_2\text{Mg}_5(\text{Si}_4\text{O}_{11})_2(\text{OH})_2$. The single and double chain silicates are collectively known as Inosilicates.

⇒ The sheet or phyllo silicates have the general formula $(\text{Si}_2\text{O}_5)_n^{2n-}$. Each SiO_4 tetrahedron shares three oxygen atoms with others thereby forming 2-D or two dimensional sheets. The layers are held together by weak Van der Waals forces. They can be easily cleaved just like graphite.

Common examples of this type of silicates include:
 1/ Talc - $\text{Mg}_3\text{Si}_2\text{O}_5(\text{OH})_2$. A soft material with smooth and greasy touch.

2/ Micas - with the general formula: $X_2Y_4Z_8\text{O}_{20}(\text{OH}, \text{F})_4$.
 where; $X = \text{K}, \text{Na}$ or Ca .

$Y = \text{Al}, \text{Mg}$ or Fe
 $Z = \text{Si}$ or Al

Typical examples are: muscovite mica - represented as $-\text{KAl}_2(\text{AlSi}_3\text{O}_{10})(\text{F}, \text{OH})_2$. And the second one is Lepidolite represented as $-\text{KLi}_2\text{Al}(\text{Al}_5\text{Si})_3\text{O}_{10}(\text{F}, \text{OH})_2$.

3/ Clay - It is a sheet structured silicate with ~~as known~~ without a known chemical formula.

4/ Kaolinite - $\text{AlSiO}_5(\text{OH})_5$.

⇒ The best in the classification of the silicates is the 3-D / three dimensional or Tecto silicates with the general formula $(\text{SiO}_2)_n$.

All the oxygen atoms of 3 SiO_4 are shared with other tetrahedra and by so doing form a three-dimensional network or framework of silicates.

Notable examples are:

Quartz (SiO_2), Tridymite and Cristobalite which are both crystalline forms of silica. When the SiO_4 units are replaced by AlO_4^{5-} units, three dimensional (3D) aluminosilicates are formed. Others are: Feldspar, Zeolites, etc.

— NB: Silicate minerals are very common in the earth crust largely because their primary units Si and O are the most abundant elements.

— Basic silicate minerals react with weak acids to undergo weathering.

— Occurrence of silicates.

— Methods of preparation of silicates.

↳ Assignment ←

* In not less than 300 words each, write comprehensive essay on:

① How do silicates occur in nature?

② Outline 4 methods of preparation of silicates.

— submit on ~~for~~ March 07/24.